



Assignment 1

Numerical Analysis

Solve all the given problems:

- (a) Apply the Euler-Cauchy method with step size h to the IVP

$$y' = -y, \quad y(0) = 1.$$

Now determine an explicit expression for y_n . For what values of h , the sequence $\{y_n\}$ is bounded?

- (b) The R-K method of the form is given by

$$y_{n+1} = y_n + h (w_1 K_1 + w_2 K_2),$$

where $K_1 = f(y_n)$ and $K_2 = f(y_n + a h (K_1 + K_2))$ for the solution of the IVP $y' = f(y)$, $y(x_0) = y_0$. Therefore, find the values of a , w_1 , and w_2 , and discuss the order of the method.

- (c) The iteration method defined by $x_{k+1} = \frac{1}{2}(x_k^2 + c)$, where $0 < c < 1$, has two fixed points ξ_1, ξ_2 , where $0 < \xi_1 < 1 < \xi_2$.

Show that $x_{k+1} - \xi_1 = \frac{1}{2}(x_k + \xi_1)(x_k - \xi_1)$, $k = 0, 1, 2, \dots$, and deduce that

$\lim_{k \rightarrow \infty} x_k = \xi_1$ if $0 \leq x_0 < \xi_2$.